

DATA LOAD

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Data load karein
df = pd.read_csv("student-mat.csv", sep=';')

# Pehli 5 rows dekhne ke liye
print(df.head())
```

```
   school sex  age address famsize Pstatus  Medu  Fedu   Mjob   Fjob   ... \
0      GP   F   18      U    GT3      A     4     4  at_home  teacher  ...
1      GP   F   17      U    GT3      T     1     1  at_home   other  ...
2      GP   F   15      U   LE3      T     1     1  at_home   other  ...
3      GP   F   15      U    GT3      T     4     2  health  services  ...
4      GP   F   16      U    GT3      T     3     3   other    other  ...
```

```
   famrel freetime  goout  Dalc  Walc  health  absences  G1  G2  G3
0      4         3      4      1      1      3         6   5   6   6
1      5         3      3      1      1      3         4   5   5   6
2      4         3      2      2      3      3        10   7   8  10
3      3         2      2      1      1      5         2  15  14  15
4      4         3      2      1      2      5         4   6  10  10
```

```
[5 rows x 33 columns]
```

Data Cleaning

```
# Missing values check karein
print(df.isnull().sum())

# Duplicates check karein
print(f"Duplicates found: {df.duplicated().sum()}")
```

```
school      0
sex         0
age         0
address     0
famsize     0
Pstatus     0
Medu        0
Fedu        0
Mjob        0
Fjob        0
reason      0
guardian    0
traveltime  0
studytime   0
failures    0
schoolsup   0
famsup      0
paid        0
activities  0
nursery     0
higher      0
internet    0
romantic    0
famrel      0
freetime    0
goout       0
Dalc        0
Walc        0
health      0
absences    0
G1          0
G2          0
G3          0
dtype: int64
Duplicates found: 0
```

ANALYSIS

```
# 1. Average Final Grade (G3)
avg_g3 = df['G3'].mean()
print(f"Average Final Grade: {avg_g3:.2f}")

# 2. Kitne students ne 15 se zyada score kiya?
```

```
high_scorers = df[df['G3'] > 15].shape[0]
print(f"Students scoring above 15: {high_scorers}")
```

```
# 3. Gender wise performance
gender_performance = df.groupby('sex')['G3'].mean()
print("\nAverage score by Gender:")
print(gender_performance)
```

Average Final Grade: 10.42
Students scoring above 15: 40

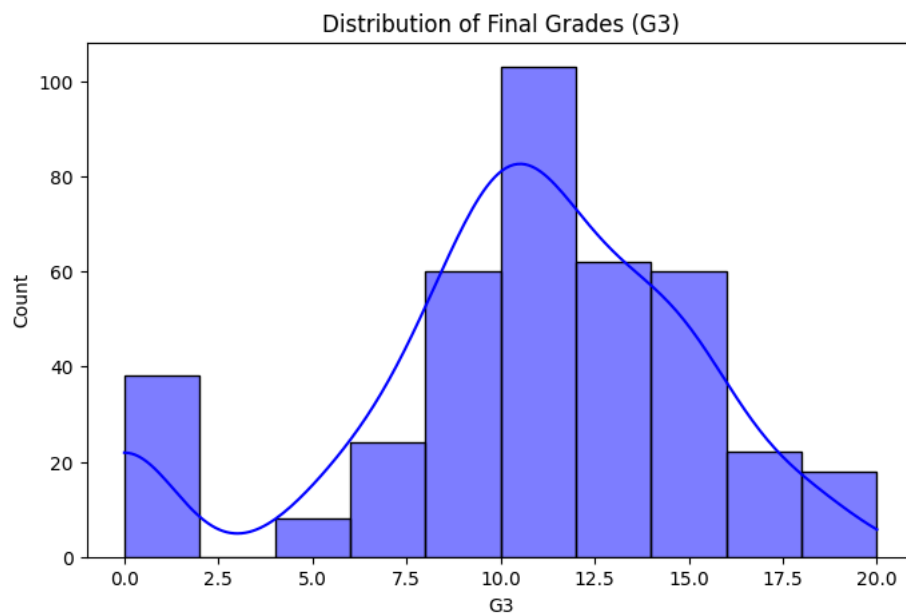
Average score by Gender:

sex	G3
F	9.966346
M	10.914439

Name: G3, dtype: float64

VISUALISATION (A).Distribution of Final Grades

```
plt.figure(figsize=(8, 5))
sns.histplot(df['G3'], bins=10, kde=True, color='blue')
plt.title('Distribution of Final Grades (G3)')
plt.show()
```



Double-click (or enter) to edit

VISUALISATION (B).Study Time VS Final Grades

```
plt.figure(figsize=(8, 5))
sns.scatterplot(x='studytime', y='G3', data=df)
plt.title('Study Time vs Final Grades')
plt.show()
```

Study Time vs Final Grades

